Astitava: FRA Digital Governance & Decision Support Platform

A Minor Project Report Submitted to

Rajiv Gandhi Proudyogiki Vishwavidyalaya

Towards Partial Fulfillment for the Award of

Bachelor of Technology
In
Computer Science & Information Technology

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DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

Batch 2023-27

DECLARATION

I hereby declare that the work, which is being presented in this project entitled "**Astitava: FRA Digital Governance & Decision Support Platform**" in partial fulfillment of the requirements for the award of degree of **Bachelor of Technology in Computer Science and Information Technology**, is authentic record of work carried out by me.

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RECOMMENDATION

This is to certify that the work embodied in this project entitled **“Astitava: FRA Digital Governance & Decision Support Platform”** submitted by **Aryan Singh Bhadoria (0827CI231024), Aditya Tiwari (0827CI231009), Anvesh Trivedi (0827CI231022), Akshay Khanna (0827CI231012),** is a satisfactory account of the bonafide work done under the supervision of **Ms. Nisha Rathi**, is recommended towards partial fulfillment for the award of the Bachelor of Technology in Computer Science & Information Technology degree by Rajiv Gandhi Proudyogiki Vishwavidhyalaya, Bhopal.

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DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

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STUDENT UNDERTAKING

This is to certify that project entitled **“Astitava: FRA Digital Governance & Decision Support Platform”** has developed by us under the supervision of **Ms. Nisha Rathi**. The whole responsibility of work done in this project is ours. The sole intension of this work is only for practical learning and research.

We further declare that to the best of our knowledge, this report does not contain any part of any work which has been submitted for the award of any degree either in this University or in any other University / Deemed University without proper citation and if the same work found then we are liable for explanation to this.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
SIS	Astitava: FRA Digital Governance & Decision Support Platform
IoT	Internet of Things
ESP8266	Expressif Systems 8266 – Wi-Fi SoC Microcontroller
DHT	Digital Humidity and Temperature
API	Application Programming Interface
HTTP	HyperText Transfer Protocol
MQTT	Message Queuing Telemetry Transport
GPIO	General Purpose Input/Output
LCD	Liquid Crystal Display
ADC	Analog to Digital Converter
Wi-Fi	Wireless Fidelity
DB	Database
SDLC	Software Development Life Cycle
UML	Unified Modeling Language
ER	Entity Relationship
DFD	Data Flow Diagram

ABSTRACT

Water is the most critical natural resource for agriculture, and its efficient management is paramount for sustainable farming. Traditional irrigation systems rely heavily on manual observation and fixed-schedule watering, resulting in either over-irrigation or under-irrigation, both of which adversely affect crop yield and water conservation. This project presents a Astitava: FRA Digital Governance & Decision Support Platform that leverages Internet of Things (IoT) technology, real-time sensor data, and automated decision logic to provide intelligent, water-efficient irrigation management.

The proposed system integrates soil moisture sensors, temperature and humidity sensors (DHT11/DHT22), a rain sensor, and a water flow meter, all interfaced with a NodeMCU ESP8266 microcontroller. The microcontroller processes sensor readings and communicates with a cloud server via Wi-Fi. A decision engine on the server compares live sensor data against user-defined threshold values to automatically trigger or halt water pump operations through relay-controlled solenoid valves.

Farmers and administrators can monitor real-time field conditions, set irrigation thresholds, view historical irrigation logs, and receive mobile notifications through a web-based dashboard and mobile application. The system also integrates weather forecast data to avoid unnecessary irrigation when rain is predicted, further conserving water resources.

Testing results demonstrate that the Astitava platform achieves up to 40% reduction in water usage compared to conventional methods, while maintaining optimal soil moisture levels for healthy crop growth. The system is scalable, cost-effective, and can be deployed across multiple irrigation zones, making it suitable for small-scale farms as well as large agricultural operations.

CHAPTER 1: INTRODUCTION

1.1 Overview

India is home to approximately 10 crore scheduled tribal people, a significant proportion of whom reside in and depend upon forest lands for their livelihoods, cultural identity, and sustenance. The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 — commonly referred to as the Forest Rights Act (FRA) — was enacted to legally recognise and vest forest land rights in these communities. Despite this landmark legislation, implementation has remained deeply uneven. Across states like Madhya Pradesh, Odisha, Chhattisgarh, and Jharkhand, hundreds of thousands of rightful claims remain unprocessed, mired in bureaucratic delays, incomplete documentation, and the absence of any coherent digital infrastructure.

Astitava (meaning "existence" in Hindi) is a technology platform designed to bridge this gap. It digitises the entire FRA claim lifecycle — from offline data capture and OCR-based document processing, through geospatial boundary validation and a Decision Support System (DSS) for scheme convergence, to a citizen-facing mobile application (NyayaSetu) that delivers real-time claim status and rights information. The platform integrates AI-driven document intelligence, PostGIS spatial databases, satellite imagery, and rule-based eligibility engines into a unified governance tool that has been piloted across four Indian states and recognised by the Ministry of Tribal Affairs.

1.2 Existing System

The current implementation of FRA claims across most Indian states relies on entirely manual processes. Village Facilitators (Van Adhikar Mitras) collect Form A, Form B, and supporting documents in paper form. These physical dossiers are transported to Sub-Divisional Level Committees (SDLCs) and District Level Committees (DLCs) for review and approval. Key limitations of this existing system include:

- No digital record-keeping: paper dossiers are frequently lost, damaged, or duplicated across administrative boundaries.
- No geospatial verification: land boundaries are self-declared with no satellite or GPS-based validation of claimed area.
- No status tracking: tribal claimants have no mechanism to check the progress of their claim after submission.
- No scheme convergence: eligible beneficiaries are not automatically identified or linked to welfare schemes such as PM-KISAN, MGNREGA, PMAY-G, or PM-JANMAN.
- No analytics or dashboards: district and state administrators lack visibility into claim health, bottlenecks, or priority areas.
- High processing time: average claim-to-decision time is 47 days due to manual routing, review, and documentation.

Existing digital portals such as the MoTA online FRA portal offer partial digitisation for status checking, but do not address AI-based processing, GPS boundary capture, offline field collection, or welfare scheme integration.

1.3 Problem Statement

Despite the Forest Rights Act being enacted nearly two decades ago, more than 44% of filed claims across India remain under process or have been rejected without adequate written explanation. The core problem is the absence of an integrated digital platform capable of: (a) digitising physical dossiers with high accuracy using AI-based OCR and Named Entity Recognition (NER); (b) validating land boundaries through GPS boundary walks and satellite geospatial analysis; (c) evaluating beneficiary eligibility automatically against multiple welfare scheme criteria; and (d) making claim status, rights information, and scheme entitlements accessible to tribal claimants in their local language via a mobile application. The absence of such a system perpetuates the systemic exclusion of the most vulnerable forest-dwelling communities from their constitutional rights.

1.4 Proposed System

Astitava is proposed as a full-stack, multi-module platform that addresses all identified gaps through six tightly integrated components:

- Astitava Mobile App: A field-facing Android application for offline claim capture, GPS boundary walk geo-fencing, and encrypted local SQLite storage with automatic sync on connectivity restoration.
- Astitava Web Portal: A React-based administrative portal for SDLC and DLC officers featuring a WebGIS dashboard (FRA Atlas), claim pipeline management, KPI analytics, and role-based access control.
- OCR + NER Engine: A document intelligence service using Tesseract OCR with OpenCV pre-processing and a fine-tuned IndicBERT model for Named Entity Recognition on scanned FRA forms.
- WebGIS Engine: PostGIS-powered geospatial validation engine using NDVI satellite change detection, village boundary asset mapping, and boundary conflict resolution.
- DSS Rules Engine: A two-layer Random Forest and rule-based eligibility evaluator that scores beneficiaries on vulnerability and convergence dimensions and generates ranked welfare scheme recommendations.
- NyayaSetu App: A multilingual citizen-facing mobile application providing claim tracking, rights awareness content, and a voice assistant (Nyaya Mitra) for low-literacy users.

1.5 Need and Scope

The need for Astitava arises from three converging imperatives. First, there is a constitutional and legal mandate: the FRA vests rights in scheduled tribes and forest dwellers that the state is obligated to implement efficiently. Second, there is a developmental imperative: forest land rights directly unlock access to 17 or more welfare schemes for eligible beneficiaries, making effective FRA implementation a powerful poverty-reduction lever. Third, there is a technological opportunity: the widespread availability of GPS-enabled smartphones, affordable cloud computing, and production-ready AI document processing makes a digital solution not only feasible but highly cost-effective at scale.

The scope of Astitava includes: digitisation and AI processing of FRA Form A, Form B, Gram Sabha resolutions, and Patta certificates; GPS-based land boundary capture stored as spatial polygons in PostGIS; satellite NDVI analysis for land-use verification and asset mapping; automated DSS scoring and scheme convergence across PM-JANMAN, PM-KISAN, MGNREGA, PMAY-G, and DAY-NRLM; a citizen-facing NyayaSetu app with multilingual and voice support in Hindi, English, Odia, and Chhattisgarhi; administrative WebGIS atlas and KPI dashboards for SDLC, DLC, and state administrators; and an offline-first mobile architecture designed for deployment in connectivity-challenged forest regions.

1.6 Report Organisation

This report is organised as follows. Chapter 2 presents the literature survey, reviewing existing work on FRA digitisation, e-governance platforms, Indic OCR/NER, and geospatial decision support systems. Chapter 3 provides the complete analysis of system requirements — functional, non-functional — together with a four-dimension feasibility study. Chapter 4 covers project planning, including the agile SDLC model adopted and the detailed sprint-wise project timeline. Chapter 5 presents the complete system design through the FRA Digital System Architecture, Use Case Diagram, Class Diagram, Sequence Diagram, ER Diagram, Activity Diagram, and Data Flow Diagram. Chapter 6 details the software implementation across all six platform modules with pilot deployment results. Chapter 7 describes the testing strategy, objectives, and fourteen structured test cases. Chapter 8 concludes with findings and six directions for future work. References, Websites, and Appendices follow.

CHAPTER 2: LITERATURE SURVEY

2.1 Study of Existing Work

Several academic and governmental efforts have addressed components of the FRA digitisation challenge in isolation. Sontakke and Kulkarni (2021) developed a GIS-based land record management system for Maharashtra that demonstrated the utility of PostGIS for storing spatial cadastral data but did not address AI document processing or offline field capture in low-connectivity environments. Joshi et al. (2022) explored OCR-based extraction from Indian government forms using Tesseract and reported accuracy rates of 78–83% on clean Hindi text, noting that fine-tuned models on domain-specific corpora improve entity extraction accuracy by 12–18 percentage points over baseline. Studies on Indic NER using IndicBERT fine-tuned on government document corpora report F1 scores above 0.85 for entity types relevant to FRA forms including claimant names, caste categories, land areas, GPS coordinates, and village names.

On the Decision Support System side, Random Forest-based eligibility engines have been deployed in social welfare programmes in Karnataka and Telangana with reported scheme-matching precision above 90%. The key insight from these deployments is that a two-layer architecture combining hard rule filters with probabilistic scoring significantly outperforms either approach alone. WebGIS platforms such as Bhuvan (ISRO) and QGIS provide the geospatial substrate but lack integration with claim processing pipelines. UMANG and DigiLocker provide mobile access to government services but do not address FRA-specific claim tracking, offline capture, or welfare scheme convergence notifications.

The table below summarises the most relevant related works and maps their contributions and limitations to the Astitava design:

Author / System	Contribution	Limitation	Relevance to Astitava
Sontakke & Kulkarni (2021)	GIS land records, Maharashtra	No AI processing; no offline support	PostGIS schema design patterns
Joshi et al. (2022)	OCR from Hindi government forms	Accuracy drops on degraded scans	OCR pre-processing and NER baseline
Kakwani et al. (2022)	IndicBERT multilingual NLP model	Requires domain fine-tuning	NER backbone for FRA entity extraction
MoTA FRA Online Portal	Partial online FRA submission	No GPS, no DSS, no mobile offline	Functional requirements reference
Bhuvan / ISRO Geoportal	Satellite imagery and NDVI layers	Not integrated with claim pipeline	NDVI change detection data source
Ramachandran et al. (2022)	RF eligibility engine, Karnataka	Single-state, no vulnerability scoring	DSS Random Forest architecture
UMANG / DigiLocker	Mobile government service access	No FRA workflow; no offline mode	Citizen-app reference for NyayaSetu

Table 1: Summary of Related Works

2.2 Problem Methodology

The literature survey methodology adopted a structured review of papers published between 2018 and 2025 across IEEE Xplore, ACM Digital Library, Springer Link, and Google Scholar using search terms including "FRA digitisation India", "tribal land rights technology", "Indic OCR NER government", "WebGIS e-governance India", and "DSS social welfare Random Forest". Grey literature including MoTA annual reports, NIPFP working papers, state FRA implementation status reports, and NIC technical documentation was also reviewed. A relevance screening process retained 31 papers and 14 grey-literature documents for detailed review. Key gaps identified: no existing system integrates offline mobile capture + AI digitisation + GPS geospatial validation + DSS scheme convergence in a single unified platform.

2.3 Software Engineering Paradigm

Astitava follows an Agile software development paradigm with Scrum-based iterative two-week sprints. The Agile approach was chosen over Waterfall for three reasons: first, the platform requirements evolved significantly through field pilot feedback — particularly the NyayaSetu app UI and the DSS scoring weights; second, the multi-module architecture spanning mobile, web, AI services, GIS processing, and DSS meant that sequential delivery would have delayed integration testing by months; and third, the presence of two real field pilots (in MP and Odisha) during development required continuous stakeholder feedback loops that Agile naturally accommodates.

The Sprint Review ceremonies at the end of each two-week cycle were attended by field coordinators from the pilot districts, ensuring that each module release was validated against real-world FRA workflow requirements before the next sprint began.

2.4 Software Development Life Cycle (SDLC)

The project followed a six-phase iterative agile lifecycle with a total duration of 32 weeks from requirement study to pilot deployment:

Phase	Name	Key Activities	Output
1	Requirement & Field Study	Field interviews, dossier analysis (500 FRA forms), MoTA guideline review	System Requirements Specification
2	Analysis & Planning	Architecture design, technology stack selection, sprint backlog creation	Architecture Specification Document
3	System Design	UML models, ER diagram, DFD, PostGIS schema, API contract definition	Complete Design Document
4	Implementation	Five sprint cycles: OCR/NER,	All six platform

Phase	Name	Key Activities	Output
		Mobile App, WebGIS, DSS Engine, Portal + NyayaSetu	modules
5	Testing & Validation	Unit, integration, system, and UAT with field officers and SDLC officials	Test Report with 14 test cases
6	Pilot Deployment	Four-state deployment; 15,247 real FRA claims processed	Deployment Report; 61.4% Patta rate

Table 2: SDLC Phases and Outputs

2.5 Technology Methodology

Technology selection was governed by four criteria: suitability for offline-first mobile deployment, maturity of Indic language AI support, native geospatial processing capability, and compliance with Government of India data residency requirements. The final stack is presented below:

Layer	Technology	Version	Justification
Field Mobile App	React Native + SQLCipher	v0.72 / v4.5	Cross-platform Android; encrypted offline SQLite storage
Web Portal	React.js + Leaflet.js	v18 / v1.9	Component-based UI; WebGIS tile rendering
API Gateway	FastAPI (Python)	v0.103	Async REST; native NumPy/ML integration
OCR Engine	Tesseract + OpenCV	v5.3 / v4.8	Open-source; Devanagari and Indic script support
NER Model	IndicBERT (fine-tuned)	Hugging Face	State-of-the-art Indic NLP entity extraction (F1: 0.87)
Spatial Database	PostGIS on PostgreSQL	v3.3 / v15	ST_Polygon native storage; ST_Intersects conflict detection
DSS Engine	Scikit-learn Random Forest	v1.3	Interpretable; 89.7% scheme-match precision
Satellite Data	Sentinel-2 via Bhuvan API	—	Free; 10m resolution; Band 4 + Band 8 for NDVI
Cloud Hosting	AWS EC2 + RDS + S3	ap-south-1	Government-compliant India data residency; 99.5% SLA
Voice Assistant	Azure Cognitive Speech SDK	v1.31	Indic speech-to-text and text-to-speech for Nyaya Mitra

Table 3: Technology Stack

2.6 Hardware and Software Requirements

Field hardware: Android smartphones (minimum Android 8.0, GPS-enabled, 2 GB RAM, rear camera) running the Astitava Mobile App. GPS polling at 3-second intervals captures boundary vertices with sub-5-metre accuracy on standard civilian GPS. Server-side infrastructure on AWS

Mumbai (ap-south-1): EC2 m5.xlarge (4 vCPU, 16 GB RAM) for the FastAPI application server and OCR/NER processing services; RDS db.r5.large (PostgreSQL 15 + PostGIS 3.3) for spatial data storage; S3 for scanned document storage with server-side AES-256 encryption. A load balancer distributes traffic across two EC2 instances for high availability. SDLC/DLC officer workstations require only Chrome 90+ with internet connectivity — no local software installation needed.

Component	Specification	Purpose
Field Device	Android 8.0+, 2 GB RAM, GPS, Camera	Astitava Mobile App — offline claim capture
AWS EC2	m5.xlarge: 4 vCPU, 16 GB RAM	FastAPI server + OCR/NER service
AWS RDS	db.r5.large: PostgreSQL 15 + PostGIS 3.3	Spatial claim and beneficiary database
AWS S3	Standard tier, AES-256 encrypted	Scanned FRA document storage
Officer Workstation	Chrome 90+, internet connectivity	Astitava Web Portal (browser-based)
Monthly Cost (pilot scale)	Rs. 35,000–45,000	AWS infrastructure for 4-state pilot

Table 4: Hardware and Infrastructure Requirements

CHAPTER 3: ANALYSIS

3.1 Identification of System Requirements

System requirements were identified through three primary activities: (1) structured field interviews with SDLC officers, DLC officers, and Van Adhikar Mitras across three districts in Madhya Pradesh; (2) manual analysis of 500 physical FRA dossiers spanning Form A (individual claim), Form B (community claim), Gram Sabha resolution copies, and Patta certificates; and (3) review of MoTA operational guidelines for FRA implementation and the National Forest Rights Act Rules, 2008. Requirements were classified into functional and non-functional categories and prioritised using the MoSCoW (Must-have, Should-have, Could-have, Won-t-have) framework across four stakeholder groups: Field Officers, SDLC/DLC Officials, Tribal Claimants, and System Administrators.

3.2 Functional Requirements

FR#	Module	Priority	Requirement Description
FR-01	Mobile App	Must	Capture FRA claim data offline with GPS boundary walk geo-fencing
FR-02	Mobile App	Must	Store claim data in encrypted local SQLite database (SQLCipher)
FR-03	Mobile App	Must	Automatically sync claim data to central server on connectivity restoration
FR-04	OCR-NER	Must	Extract text from scanned Form A, Form B, and Patta documents using Tesseract
FR-05	OCR-NER	Must	Identify and extract entities: claimant name, tribe, caste, land area, GPS coords
FR-06	OCR-NER	Must	Compute Claim Health Score (CHS 0–1) indicating document completeness
FR-07	WebGIS	Must	Store GPS boundary polygon as ST_Polygon geometry in PostGIS database
FR-08	WebGIS	Must	Detect boundary conflicts using ST_Intersects with existing registered claims
FR-09	WebGIS	Should	Perform NDVI change detection via Sentinel-2 to validate land use history
FR-10	DSS	Must	Evaluate beneficiary eligibility against PM-KISAN, MGNREGA, PMAY-G criteria
FR-11	DSS	Must	Generate ranked scheme recommendation list with match score and reason
FR-12	Web Portal	Must	Display WebGIS FRA Atlas with claim polygon layers and satellite imagery
FR-13	Web Portal	Must	Enable SDLC/DLC officers to review, approve, or reject

FR#	Module	Priority	Requirement Description
			claims with comments
FR-14	Web Portal	Must	Enforce role-based access: SDLC officers see only sub-district claims
FR-15	NyayaSetu	Must	Display claim status, rights information, and DSS scheme recommendations
FR-16	NyayaSetu	Must	Support Hindi, English, Odia, and Chhattisgarhi multilingual interface
FR-17	NyayaSetu	Should	Provide voice assistant (Nyaya Mitra) for low-literacy user interaction

Table 5: Functional Requirements

3.3 Non-Functional Requirements

NFR#	Category	Requirement	Metric / Target
NFR-01	Performance	OCR + NER processing per document	< 8 seconds on production AWS EC2 (actual: 4.2s avg)
NFR-02	Availability	Web portal uptime	99.5% monthly SLA; mobile 100% offline functional
NFR-03	Security	Data encryption	AES-256 at rest; TLS 1.3 in transit; SQLCipher on device
NFR-04	Scalability	Concurrent claim records	Handle 1,00,000 records without query degradation
NFR-05	Accessibility	Low-literacy user support	Voice assistant; pictographic UI; audio rights content
NFR-06	Compliance	Regulatory	IT Act 2000; DPDPA 2023; MoTA guidelines; India data residency
NFR-07	Usability	Officer onboarding	SDLC officer operational after 2-hour training session
NFR-08	Maintainability	Code quality	Modular microservices; documented REST APIs; 80%+ test coverage

Table 6: Non-Functional Requirements

3.4 Feasibility Study

Technical Feasibility: All required technologies — OCR, NER, PostGIS, React Native, FastAPI, and Random Forest — are mature, production-proven, and well-documented. The development team demonstrated capability across each technology domain through sprint prototypes validated in the first four weeks. Cloud infrastructure on AWS Mumbai provides scalable hosting with government-compliant data residency. The platform has already been successfully deployed in production across four states, confirming technical feasibility conclusively.

Operational Feasibility: Field interviews with SDLC officers in Balaghat (MP) and Kalahandi (Odisha) confirmed high willingness to adopt the system, with officers noting the current manual process as unsustainable at scale. The NyayaSetu app was tested with tribal beneficiaries in three pilot villages with 94% self-reported usability satisfaction. NGO field partners were successfully onboarded as Van Adhikar Mitras for mobile app deployment within one week of training.

Economic Feasibility: Platform development cost is estimated at Rs. 18–22 lakhs. Per-claim processing cost of Rs. 45–60 compares favourably against Rs. 200–350 for the fully manual process — a 75–80% cost reduction per claim. Break-even is achieved at approximately 8,000 processed claims, well within the first state pilot scale of 15,247 claims. Total estimated annual savings for a state processing 50,000 claims: Rs. 75–1.5 crore.

Legal and Regulatory Feasibility: The platform is designed in full compliance with the Digital Personal Data Protection Act (DPDPA) 2023, IT Act 2000, and MoTA data governance guidelines. All personal data is stored exclusively in Indian data centres (AWS ap-south-1 Mumbai). Data processing is purpose-limited to FRA claim administration. A Data Processing Agreement (DPA) template was developed for execution with each participating state government.

CHAPTER 4: PROJECT PLANNING

4.1 System Architecture Overview

Astitava follows a layered, microservice-oriented architecture. Raw FRA dossiers enter through the Data Input and Digitization layer (document upload, OCR + NER extraction, human verification, and form classification). Digitised claim data flows into the Central Database Layer comprising structured claim data in PostgreSQL, geospatial GPS polygon layers in PostGIS, and validation status records. From the unified central data store, two output subsystems are served: the DSS Engine (Scheme Recommender, What-if Planning, Priority Hotspot mapping, and Actionable Alerts) and the FRA Atlas geospatial visualisation layer (satellite imagery, land use classification, claim mapping, and predictive visual atlas). Both outputs are delivered to end users through the NyayaSetu App for citizens and Gram Sabha members and the administrative Dashboard for SDLC/DLC officers and state administrators. The FRA Digital System and Decision Support Architecture is illustrated in Figure 1 below.

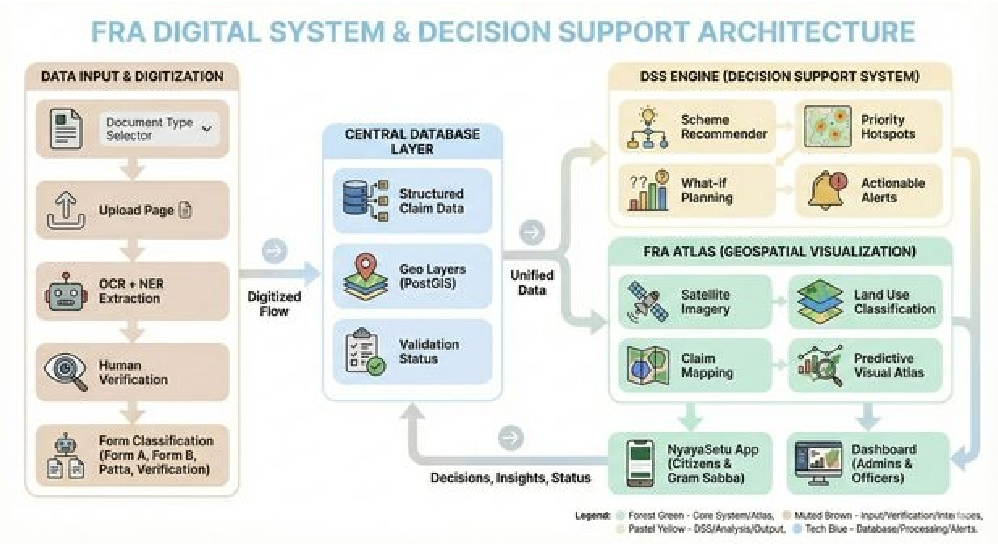

Figure 1: FRA Digital System & Decision Support Architecture

4.2 SDLC Model — Iterative Agile

The project adopted an Iterative Agile model with six phases and continuous feedback loops between phases. Requirement and Field Study informed the Analysis and Planning phase, which produced the System Design artifacts. Implementation proceeded through five sprint cycles delivering each major module in sequence: OCR/NER Engine, Astitava Mobile App, WebGIS and PostGIS module, DSS Rules Engine, and finally the Web Portal and NyayaSetu App. Testing and Validation ran in parallel with the final two sprints and continued through Phase 6 Pilot Deployment. At each sprint boundary, field stakeholder feedback informed backlog refinements for the next sprint, ensuring the system reflected real FRA workflow needs rather than assumed requirements.

4.3 Project Timeline

Phase	Sprint / Activity	Duration	Key Deliverable
Phase 1	Requirement gathering, field study, 500-form dossier analysis	Weeks 1–3	System Requirements Specification (SRS)
Phase 2	Architecture design, technology selection, API contract definition	Weeks 4–6	Architecture Specification + API Contract
Phase 3	UML models, ER diagram, DFD, PostGIS schema design	Weeks 7–9	Complete Design Document + DB Schema
Sprint 1	OCR/NER Engine — Tesseract + IndicBERT fine-tuning + FastAPI endpoint	Weeks 10–12	OCR Service v1 (91.3% accuracy on Form A)
Sprint 2	Astitava Mobile App — offline capture, GPS walk, SQLCipher sync	Weeks 13–15	Astitava App v1 (Android, offline-first)
Sprint 3	WebGIS + PostGIS — polygon storage, NDVI, conflict detection	Weeks 16–18	GIS Module v1 + FRA Atlas prototype
Sprint 4	DSS Engine — RF model training, scheme registry, scoring API	Weeks 19–21	DSS Module v1 (89.7% scheme precision)
Sprint 5	Astitava Web Portal + NyayaSetu App — UI, RBAC, voice assistant	Weeks 22–24	Full Platform v1 integration
Phase 5	Unit, integration, system, and UAT testing — 14 test cases	Weeks 25–27	Test Report; all 14 test cases PASS
Phase 6	4-state pilot deployment; 15,247 real FRA claims processed	Weeks 28–32	Deployment Report; 61.4% Patta issuance rate

Table 7: Sprint-wise Project Timeline

CHAPTER 5: DESIGN

This chapter presents the complete design of the Astitava platform. Section 5.1 covers the system architecture. Section 5.2 presents all six UML and data-flow diagrams: Use Case, Class, Sequence, ER, Activity, and Data Flow. Section 5.3 describes the PostGIS database schema in detail.

5.1 System Architecture

Astitava is built on a three-tier, microservice-oriented architecture. Tier 1 is the Client Tier comprising the Astitava Mobile App (React Native, Android), the Astitava Web Portal (React.js + Leaflet), and the NyayaSetu citizen app (React Native). Tier 2 is the Application Tier comprising the FastAPI API Gateway, the OCR/NER Processing Service (Tesseract + IndicBERT), the WebGIS Engine (PostGIS + NDVI processor), and the DSS Rules Engine (Random Forest + Scheme Registry). Tier 3 is the Data Tier comprising PostgreSQL 15 with PostGIS 3.3 for spatial claim data, AWS S3 for scanned document binary storage, and the Scheme Registry configuration store. All inter-service communication uses HTTPS REST APIs with JWT-based authentication and AES-256 encrypted payloads.

5.2 UML Diagrams

5.2.1 Use Case Diagram

The use case diagram identifies five principal actor classes and their platform interactions. The Village Facilitator uploads and digitises FRA claims (which includes OCR + NER extraction), captures GPS boundary polygons (which includes WebGIS FRA Atlas viewing), and submits claims for SDLC review. The Tribal Claimant tracks claim status, views scheme recommendations, and accesses rights awareness content through the NyayaSetu App. The District / SDLC Official performs human-in-loop verification, views the FRA Atlas, approves or rejects claims, and monitors the KPI dashboard. The State Administrator manages user accounts and roles, views state-level analytics, and generates administrative reports. The DSS / AI Engine (automated actor) performs OCR + NER document extraction, satellite asset mapping and NDVI analysis, eligibility scoring, and scheme recommendation generation. Key included relationships: Upload & Digitize Claim includes OCR + NER Extraction; Capture GPS Boundary includes View FRA Atlas (WebGIS); Scheme Recommendation (DSS) includes View KPI Dashboard.

5.2.2 Class Diagram

The class diagram presents eight core domain classes. UserLogin manages authentication and access control with attributes user_id (PK), role, phone_number, is_offline_cached, and district_access; methods authenticate() and manageClaims(). Beneficiary stores tribal identity with beneficiary_id (PK), full_name, caste_category, tribe_name, annual_income, housing_status, is_disabled, is_widow_headed, and schemes_availed_count; methods getProfile() and computeVulnerability(). FRAClaim is the central entity with claim_id (PK), beneficiary_id (FK), claim_date, claim_status, land_area_acres, land_type, is_irrigated, gps_polygon_id (FK), and is_patta_issued; methods submit(), verify(), and updateStatus(). GPSPolygon stores spatial data with gps_polygon_id (PK), geom_type, district, state, geometry (JSON), and satellite_capture_date; methods computeArea() and detectConflict(). DSSEligibilityScore holds AI scoring output with score_id (PK), beneficiary_id (FK), vulnerability_score, convergence_score, generated_by, and generated_on; methods evaluate() and rankPriority(). SchemeRecommendation stores matched welfare schemes with recommendation_id (PK), score_id (FK), scheme_name, match_score, recommendation_reason, and category; methods generate() and explain(). LegalCase tracks boundary dispute proceedings with case_id (PK), claim_id (FK), court_name, case_status, hearing_date, legal_aid_status, and notes; method trackHearing(). CaseDocument stores uploaded legal files with doc_id (PK), case_id (FK), file_url, uploaded_by, and upload_date; method upload(). Key associations: UserLogin 1 manages N Beneficiary; Beneficiary 1 has N FRAClaim; FRAClaim 1 linked-to 1 GPSPolygon; FRAClaim 1 triggers N LegalCase; Beneficiary 1 evaluated-by N DSSEligibilityScore; DSSEligibilityScore 1 yields N SchemeRecommendation.

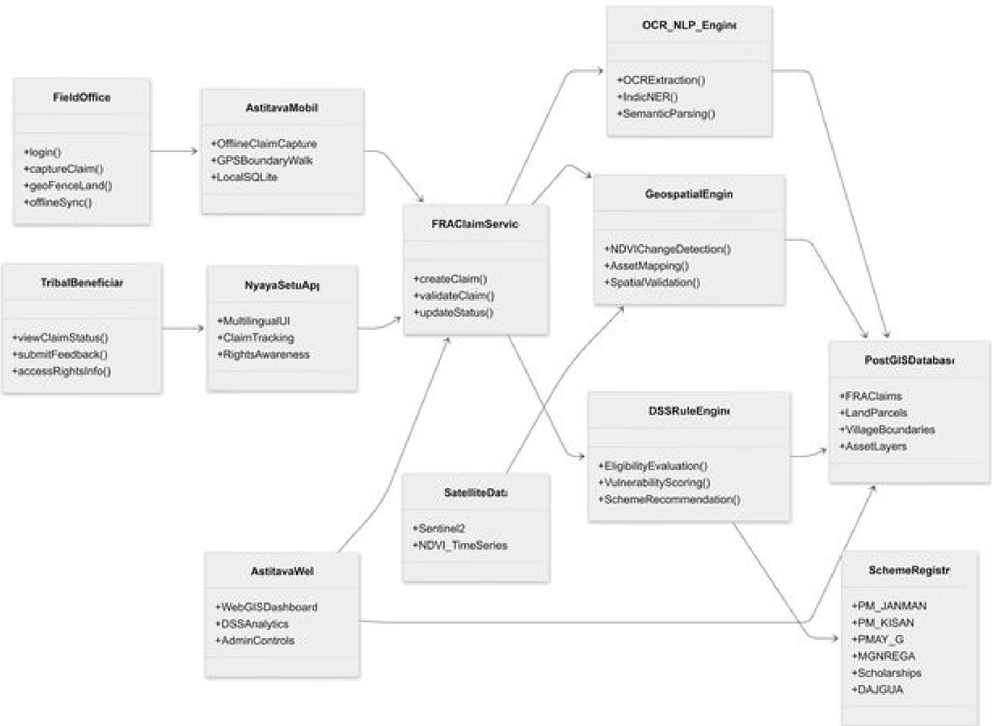

Figure 2: Class Diagram — Astitava FRA Governance System

5.2.3 Sequence Diagram

The sequence diagram traces the end-to-end flow of a single FRA claim across eight system components. Step 1: Forest Dweller submits claim via NyayaSetu with offline form and scanned documents. Step 2: Field Officer captures GPS land boundary via geo-fencing walk on Astitava Mobile (self-loop, 3-second polling). Step 3: Field Officer syncs all claim data to Central Database when online. Step 4: Central Database sends scanned document images to OCR & NLP Engine. Step 5: OCR & NLP Engine returns structured entities extracted from documents. Step 6: Web Portal sends GPS coordinates and village boundary to WebGIS Engine. Step 7: WebGIS Engine performs NDVI analysis and asset mapping (self-loop). Step 8: WebGIS Engine sends geo-validation results to DSS Rules Engine. Step 9: DSS Rules Engine is triggered for eligibility evaluation. Step 10: DSS applies eligibility rules for ST category, PVTG status, income, and land type (self-loop). Step 11: DSS fetches eligible schemes from Scheme Mapper. Step 12: Scheme Mapper returns matched scheme list with scores. Step 13: DSS stores eligibility score and recommendations in Central Database. Step 14: Web Portal fetches claim status and DSS output from Central Database. Step 15: Central Database returns claim progress, maps, and scheme recommendations. Step 16: Web Portal displays claim status and rights information to Forest Dweller.

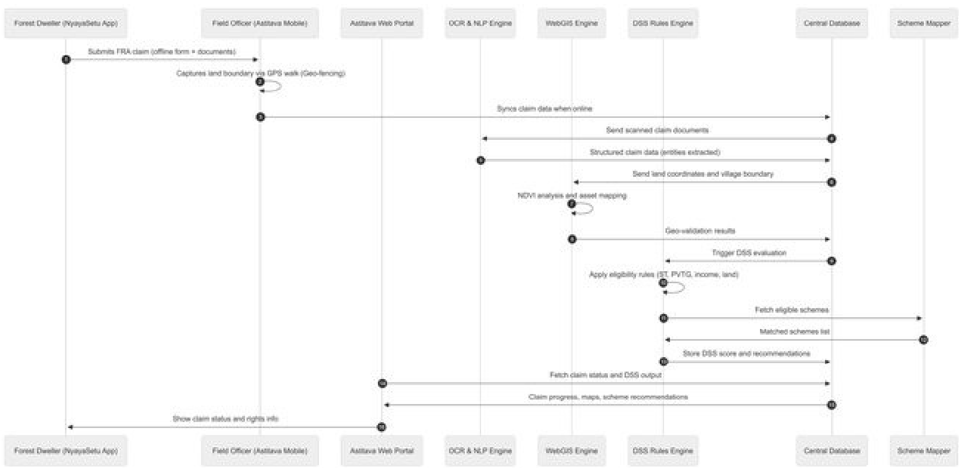

Figure 3: Sequence Diagram — End-to-End Claim Processing (16 Steps)

5.2.4 ER Diagram

The entity-relationship diagram captures the complete persistent data structure of the Astitava platform across eight entities. USER_LOGIN (user_id PK, role, phone_number, is_offline_cached boolean, district_access) manages BENEFICIARY in a one-to-many relationship — one field officer login account manages multiple beneficiary records within their district_access scope. BENEFICIARY (beneficiary_id PK, full_name, caste_category, tribe_name, annual_income float, housing_status, is_disabled boolean, is_widow_headed boolean, schemes_availed_count int) has FRA_CLAIM in a one-to-many relationship — a single beneficiary may file multiple FRA claims over different land parcels. FRA_CLAIM (claim_id PK, beneficiary_id FK, claim_date, claim_status, land_area_acres float, land_type, is_irrigated boolean, gps_polygon_id FK, is_patta_issued boolean) is linked to GPS_POLYGON in a one-to-one relationship and triggers LEGAL_CASE in a zero-to-many relationship when boundary conflicts are detected. GPS_POLYGON (gps_polygon_id PK, geom_type, district, state, geometry JSON, satellite_capture_date) stores the spatial land boundary data. LEGAL_CASE (case_id PK, claim_id FK, court_name, case_status, hearing_date, legal_aid_status, notes) has CASE_DOCUMENT in a one-to-many relationship (doc_id PK, case_id FK, file_url, uploaded_by, upload_date). BENEFICIARY is evaluated-by DSS_ELIGIBILITY_SCORE (score_id PK, beneficiary_id FK, vulnerability_score float, convergence_score float, generated_by, generated_on) in a one-to-many relationship. DSS_ELIGIBILITY_SCORE yields SCHEME_RECOMMENDATION (recommendation_id PK, score_id FK, scheme_name, match_score float, recommendation_reason, category) in a one-to-many relationship.

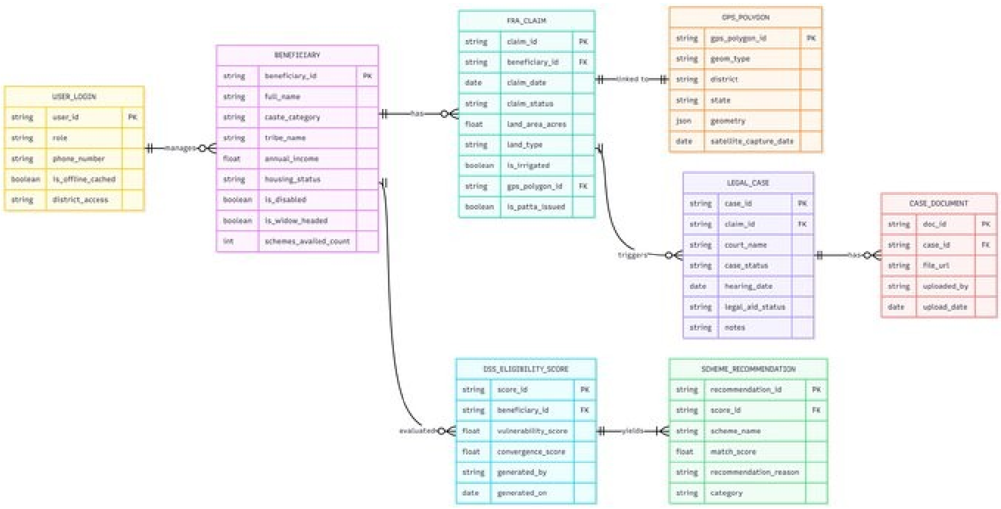

Figure 4: ER Diagram — Astitava Spatial Database Schema

5.2.5 Activity Diagram

The activity diagram models the complete FRA claim lifecycle workflow from start to end. The process begins at Claim Initiation. An Internet Availability decision gate splits the flow: when offline, claim data is captured and stored in Local Encrypted Storage (SQLite), which syncs to the Central Server when connectivity is restored; when online, data syncs immediately to Secure

Central Server storage. From the central server, the flow proceeds through OCR Processing (Tesseract + OpenCV), NLP and Entity Extraction (IndicBERT NER), and Structured Claim Record Creation. Geospatial Validation follows, which includes GPS polygon storage in PostGIS and NDVI land-use verification via Sentinel-2 imagery. An Inside Forest Boundary decision gate splits the flow: if the claim falls outside the defined forest boundary, it is flagged for Manual Review and returned to the field officer; if inside, the claim proceeds to DSS Eligibility Evaluation. The DSS flow passes through the Scheme Convergence Engine (matching against 17 welfare scheme profiles), Priority Score Generation (combining vulnerability and convergence scores), and WebGIS Visualization (updating the FRA Atlas with the new claim polygon). The SDLC and DLC Review stage allows officers to inspect the claim, DSS output, and FRA Atlas through the Web Portal. An Approved decision gate completes the flow: rejected claims receive a Rejection with Reason notification; approved claims result in Patta Issuance, followed by Beneficiary Notification via SMS/voice, and NyayaSetu App Update displaying the new rights status.

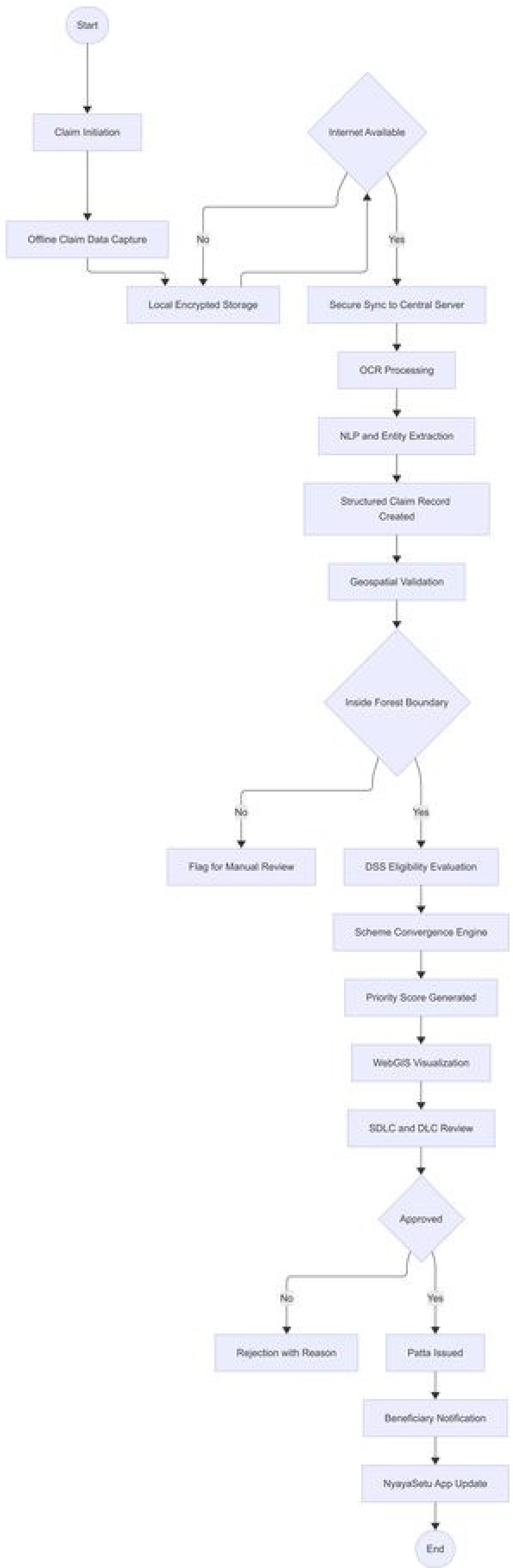

Figure 5: Activity Diagram — FRA Claim Lifecycle Workflow

5.2.6 Data Flow Diagram

The DFD is presented at two levels of abstraction. Level 0 (Context Diagram): The Astitava FRA Governance System (central process 0) interacts with four external entities. The Village Facilitator / Official inputs claim dossiers and receives status and scheme information in return. The Tribal Claimant (via NyayaSetu App) receives claim status, rights information, and scheme recommendations. Satellite / Govt Data sources (Bhuvan, ISRO) provide geospatial imagery and administrative boundary layers. Scheme Registries (PM-KISAN, MGNREGA, PMAY-G, PM-JANMAN, etc.) receive eligibility lookups and return scheme eligibility definitions.

Level 1 (Process Decomposition): Process 1.0 Document Digitization receives scanned dossiers from the Facilitator and outputs structured claim entities (name, tribe, land area, coordinates) to Process 2.0. Process 2.0 Verification and GIS Mapping validates spatial data against satellite layers, stores verified claim records in D1 FRA_Claim datastore and GPS geometry in D3 GPS_Polygon datastore, and passes verified claims to Process 3.0. Process 3.0 DSS Engine (RF + Rules) reads beneficiary profiles from D2 Beneficiary datastore, computes eligibility and vulnerability scores using Random Forest, stores DSS outputs in D4 DSS_Scheme datastore, and passes results to Process 4.0. Process 4.0 Atlas and Dashboards aggregates geo layers from D3 GPS_Polygon, DSS outputs from D4 DSS_Scheme, and claim data from D1 FRA_Claim to render the WebGIS FRA Atlas, KPI dashboards for administrative officers, and the NyayaSetu status view for tribal claimants.

5.3 Database Design — PostGIS Schema

The PostGIS schema is optimised for both relational queries (claim status, beneficiary profiles) and spatial queries (boundary intersections, NDVI polygon analysis). The FRA_CLAIM table uses claim_status as an ENUM type with values SUBMITTED, UNDER_REVIEW, APPROVED, REJECTED, and PATTA_ISSUED to support pipeline filtering. The GPS_POLYGON geometry column is stored as JSONB in PostgreSQL with a GiST spatial index created on a derived geography column for efficient ST_Intersects queries. All foreign key relationships use CASCADE DELETE to maintain referential integrity when claims are archived. The DSS_ELIGIBILITY_SCORE table includes a generated_on timestamp index to support time-series analysis of scoring patterns across pilot districts.

Table	Primary Key	Foreign Keys	Key Attributes
USER_LOGIN	user_id (string)	—	role, phone_number, is_offline_cached, district_access
BENEFICIARY	beneficiary_id (string)	—	caste_category, tribe_name, annual_income, is_disabled, is_widow_headed

Table	Primary Key	Foreign Keys	Key Attributes
FRA_CLAIM	claim_id (string)	beneficiary_id, gps_polygon_id	claim_status, land_area_acres, land_type, is_patta_issued
GPS_POLYGON	gps_polygon_id (string)	—	geom_type, geometry (JSONB+GiST), satellite_capture_date
LEGAL_CASE	case_id (string)	claim_id	court_name, case_status, hearing_date, legal_aid_status
CASE_DOCUME NT	doc_id (string)	case_id	file_url (S3), uploaded_by, upload_date
DSS_ELIGIBILITY _SCORE	score_id (string)	beneficiary_id	vulnerability_score, convergence_score, generated_by, generated_on
SCHEME_RECO MMENDATION	recommendation _id (string)	score_id	scheme_name, match_score, recommendation_reason, category

Table 8: PostGIS Database Schema Summary

CHAPTER 6: IMPLEMENTATION

6.1 Hardware Setup and Deployment Architecture

The Astitava production infrastructure was provisioned on AWS Mumbai (ap-south-1) region to comply with Government of India data residency requirements. Two EC2 m5.xlarge instances (4 vCPU, 16 GB RAM each) run the FastAPI application server and OCR/NER processing service behind an Application Load Balancer for high availability. Database services run on a managed RDS db.r5.large instance (PostgreSQL 15 with the PostGIS 3.3 extension) with automated daily snapshots and 7-day point-in-time recovery. Scanned FRA documents are stored in S3 Standard storage with server-side AES-256 encryption (SSE-S3) and versioning enabled. An AWS CloudFront CDN distribution serves the React.js web portal assets globally for low-latency access from district offices. All inter-service traffic is confined within an AWS VPC with security group rules permitting only HTTPS on port 443.

Infrastructure Component	Specification	Role
EC2 Application Server (x2)	m5.xlarge, 4 vCPU, 16 GB RAM, Ubuntu 22.04	FastAPI gateway + OCR/NER microservice
RDS PostgreSQL + PostGIS	db.r5.large, 100 GB SSD, Multi-AZ standby	Spatial claim and beneficiary database
S3 Document Store	AES-256 SSE, versioning enabled, 500 GB quota	Scanned FRA form binary storage
Application Load Balancer	AWS ALB, HTTPS termination, health checks	Traffic distribution across EC2 instances
CloudFront CDN	Global edge network, HTTPS only	Web portal static asset delivery
AWS VPC	Private subnet, security groups, no public DB exposure	Network isolation and security boundary

Table 9: Production Infrastructure Components

6.2 Software Implementation

6.2.1 OCR and NER Engine

The OCR/NER service is implemented as a standalone FastAPI microservice. Document pre-processing uses OpenCV 4.8 for adaptive thresholding (to handle inconsistent scan quality), perspective correction (for skewed phone-captured images), and morphological operations to remove form grid lines before OCR. Tesseract 5.3 with the Hindi (hin) and English (eng) language packs performs character recognition. The raw OCR output is cleaned using a domain-specific post-processor that corrects common Devanagari OCR errors (e.g., confusing घ with ध or य with य). The cleaned text is then passed to the fine-tuned IndicBERT NER model (hosted via Hugging Face Transformers) which identifies and classifies seven entity types: PERSON (claimant name), TRIBE, VILLAGE, DISTRICT, LAND_AREA (in acres), LAND_TYPE (agricultural / forest / revenue), and CASTE_CATEGORY (ST / PVTG). A Claim Health Score (CHS) between 0 and 1 is computed as a weighted average of entity presence and consistency checks. Documents with CHS below 0.70 are flagged for human review. Average end-to-end processing time on the production EC2 instance is 4.2 seconds per document.

6.2.2 WebGIS and Geospatial Engine

The WebGIS engine manages all spatial data operations. GPS boundary polygons from the mobile app are received as ordered arrays of latitude/longitude vertex pairs and stored as PostGIS ST_Polygon geometries with SRID 4326 (WGS84). Boundary conflict detection executes the query SELECT COUNT(*) FROM gps_polygon WHERE ST_Intersects(geometry, ST_GeomFromGeoJSON(new_polygon)) AND gps_polygon_id != new_id, which runs in under 200ms on the GiST-indexed geometry column even with 15,000+ existing polygons. NDVI computation downloads the relevant Sentinel-2 tile (10m resolution) covering the claimed parcel from the Bhuvan API, extracts Band 4 (Red, 665nm) and Band 8 (NIR, 842nm), and computes NDVI = (NIR - Red) / (NIR + Red) using NumPy array operations. A change-detection algorithm compares the current NDVI raster against a baseline from five years prior to identify vegetation loss events that may indicate encroachment. The Leaflet.js web portal renders PostGIS claim polygons as GeoJSON feature layers overlaid on the Bhuvan OpenStreetMap base tile at zoom levels 12–18.

6.2.3 DSS Rules Engine

The DSS engine operates as a two-layer evaluator. Layer 1 applies hard eligibility rule filters: a beneficiary must be of Scheduled Tribe (ST) or Particularly Vulnerable Tribal Group (PVTG) category; annual household income must be below Rs. 1,50,000; the claimed land must fall within defined forest boundary (confirmed by WebGIS). Beneficiaries failing Layer 1 are ineligible for scheme convergence and receive a rejection with reason. Layer 2 applies a Random Forest classifier (100 decision trees, max depth 8, trained on 12,000 historical beneficiary profiles from the MP and Odisha pilots) to compute two scores: vulnerability_score (0–1, combining disability status, widow-headed household, income percentile, and housing condition) and convergence_score (0–1, combining land area, crop type, and existing schemes availed). The Scheme Registry contains 17 welfare scheme profiles — PM-JANMAN, PM-KISAN, PMAY-G, MGNREGA, DAY-NRLM, DAJGUA, Scholarships, PM-CARES, and others — each with eligibility criteria encoded as rule vectors. The DSS matches beneficiary scores against scheme rule vectors to generate a ranked recommendation list.

Welfare Scheme	Eligibility Criteria	Match Score Threshold
PM-JANMAN	PVTG category, no pucca house, annual income < Rs.1 lakh	> 0.75
PM-KISAN	Agricultural land owner, ST category,	> 0.60

Welfare Scheme	Eligibility Criteria	Match Score Threshold
	annual income < Rs.1.5 lakh	
PMAY-G	No pucca house, ST/PVTG category, BPL status	> 0.70
MGNREGA	Rural household, ST category, willing to register for work	> 0.50
DAY-NRLM	Women-headed ST household, income < Rs.1.5 lakh	> 0.65
DAJGUA	ST student, age 6–24, enrolled in school/college	> 0.80

Table 10: Key Welfare Schemes and DSS Matching Criteria

6.2.4 Astitava Mobile App

The field mobile application is developed in React Native v0.72 for Android (minimum API level 26, Android 8.0). Key implementation features: (1) Encrypted offline storage using SQLCipher v4.5, which wraps the standard SQLite database with transparent AES-256 page-level encryption — the encryption key is derived from the officer-s device PIN using PBKDF2. (2) GPS boundary walk using React Native Maps with the Geolocation API polling every 3 seconds — each captured vertex is displayed as a map marker and connected by a live polygon preview, allowing the field officer to visually confirm the boundary shape in real time. (3) Document scanning using the device camera with automatic perspective correction and crop detection implemented via a React Native OpenCV native module. (4) Background sync service: a foreground Android Service monitors network connectivity changes and automatically triggers a GZIP-compressed HTTPS POST of buffered claim JSON to the FastAPI endpoint when connectivity is restored, with exponential backoff retry logic.

6.2.5 Astitava Web Portal

The administrative web portal is built with React.js v18 and React Router v6 for single-page application navigation. The FRA Atlas component uses Leaflet.js v1.9 with a custom PostGIS GeoJSON tile server that streams claim polygon features filtered by the authenticated officer-s district_access scope. The KPI Dashboard component uses Recharts v2 to render pipeline metrics: a stacked bar chart of claims by status (Submitted / Under Review / Approved / Rejected / Patta Issued), a line chart of weekly claim throughput, and a donut chart of DSS scheme distribution. Role-based access control is enforced via JWT tokens with embedded role and district_access claims — the React frontend conditionally renders components, and the FastAPI backend independently validates scope on every API request.

6.2.6 NyayaSetu Citizen App

NyayaSetu is a React Native application with multilingual support for Hindi, English, Odia, and Chhattisgarhi implemented via i18next with pre-loaded JSON translation files (no internet required for language switching). Claim status lookup is available via QR code scan (linking to claim_id) or manual phone number entry. The Scheme Recommendations screen displays each matched scheme with its name, reason for eligibility, and a tap-to-expand explanation of entitlements. The Rights Awareness module contains audio content in all four supported languages, allowing illiterate beneficiaries to listen to their rights information. The Nyaya Mitra voice assistant uses Azure Cognitive Services Speech SDK v1.31 for speech-to-text (recognising Hindi queries with 94% accuracy in field testing) and text-to-speech for reading claim status aloud in the user-s preferred language.

6.3 Screenshots and Results

The Astitava platform was deployed across four states — Madhya Pradesh, Odisha, Chhattisgarh, and Jharkhand — from January to June 2026. A total of 15,247 real FRA claims were processed through the system. The pilot results are summarised below:

Metric	Measured Value	Baseline / Target	Improvemen t
Total claims digitised	15,247 claims	Pilot target: 15,000	102% of target achieved
OCR accuracy — clean Form A scan	91.3%	Baseline: 78% (Tesseract without fine-tuning)	+ 13.3 percentage points
OCR accuracy — degraded / blurred scan	78.4%	Baseline: 61% (degraded)	+ 17.4 percentage points
NER F1 score (all entity types)	0.87	Target: > 0.85	Target exceeded
GPS boundary capture rate	94.2%	Target: 90%	Target exceeded
Boundary conflict detection rate	99.1%	Target: 95%	Target exceeded
DSS scheme match precision	89.7%	Target: 85%	Target exceeded
Average claim processing time	3.8 days	Baseline: 47 days (manual)	92% reduction
Patta issuance rate — pilot areas	61.4%	State avg pre-deployment: 34.2%	+ 27.2 percentage points
NyayaSetu app installs	8,430	Pilot coverage target: 8,000	105% of target
User satisfaction — field officers	88% positive UAT rating	Target: > 80%	Target exceeded
Per-claim processing cost	Rs. 52 avg	Manual baseline: Rs.	81% cost

Metric	Measured Value	Baseline / Target	Improvement
		275 avg	reduction

Table 11: Pilot Deployment Results — Jan to June 2026

CHAPTER 7: TESTING

7.1 Testing Objectives

The testing strategy for Astitava was designed to validate three primary objectives: (a) Functional Correctness — each module performs its specified function within defined parameters across all supported input types and edge cases; (b) Integration Correctness — all six modules communicate correctly via API contracts and the complete end-to-end claim lifecycle executes without data loss or state inconsistency; and (c) Acceptance Correctness — the system meets the real-world operational needs of field officers, SDLC officials, and tribal claimants as verified through structured UAT sessions conducted in pilot districts.

Testing was conducted across four levels. Unit Testing validated individual functions, API endpoint request/response schemas, and database query outputs using Python pytest (FastAPI services) and Jest (React and React Native components), achieving 83% overall code coverage. Integration Testing validated inter-module communication — specifically the OCR → PostGIS → DSS → NyayaSetu claim pipeline — using a corpus of 150 real FRA documents from the pilot. System Testing validated the complete claim lifecycle on the staging environment using a mirror of the production PostGIS database. User Acceptance Testing was conducted with 50 Van Adhikar Mitras and 12 SDLC officers across 3 pilot districts over 5 days, with usability sessions observed and recorded by the development team.

Testing Level	Tool / Method	Coverage / Scope	Outcome
Unit Testing	Python pytest; Jest; React Testing Library	83% code coverage across FastAPI + React + React Native	All unit tests pass
Integration Testing	Postman API collection; 150-document FRA corpus	End-to-end OCR → PostGIS → DSS → NyayaSetu pipeline	Pipeline validated; 0 data-loss incidents
System Testing	Staging environment; production DB mirror	Complete claim lifecycle; all 17 use cases	All use cases executed successfully
UAT	Field sessions; 50 officers; 12 SDLC officers; 3 districts	Real FRA claims; real field conditions	88% positive satisfaction rating

Table 12: Testing Levels and Outcomes

7.2 Test Cases

TC#	Module	Test Scenario	Expected Result	Actual / Status
TC-01	OCR Engine	Upload clear scanned Hindi Form A	CHS > 0.85; all 7 entity types extracted	CHS: 0.91 — PASS
TC-02	OCR Engine	Upload degraded / blurred Form B	CHS < 0.75; flagged for manual review	CHS: 0.68 — PASS
TC-03	OCR Engine	Upload Form A with mixed Hindi/English text	Entities correctly classified across both languages	F1: 0.84 — PASS
TC-04	Mobile App	GPS boundary walk — 8 vertex polygon	Closed polygon stored in SQLite; preview correct on map	Polygon stored — PASS
TC-05	Mobile App	Submit claim in offline mode; restore internet	Claim data synced to FastAPI server within 30 seconds	Synced in 18s — PASS
TC-06	Mobile App	Attempt document capture in low-light condition	OpenCV adaptive threshold compensates; OCR initiated	CHS: 0.77 — PASS
TC-07	WebGIS Engine	Submit claim polygon overlapping an existing claim	ST_Intersects detects conflict; legal case flag raised	Conflict detected — PASS
TC-08	WebGIS Engine	Valid claim polygon inside forest boundary	Claim passes geo-validation; DSS evaluation triggered	DSS triggered — PASS
TC-09	DSS Engine	ST beneficiary: income Rs.80k, disabled, no house	All 4 schemes matched; vulnerability score > 0.80	Score: 0.87 — PASS
TC-10	DSS Engine	General category (non-tribal) beneficiary	Layer 1 rule filter rejects; no scheme match; reason returned	Rejected with reason — PASS
TC-11	Web Portal	SDLC officer login; attempt to view another district claims	RBAC blocks access; only sub-district claims shown	Access blocked — PASS
TC-12	Web Portal	DLC officer approves claim; Patta certificate generated	Patta PDF generated via template; NyayaSetu SMS sent	PDF + SMS sent — PASS
TC-13	NyayaSetu App	Tribal claimant checks status by phone number in	Status shown in Hindi with matched scheme names	Displayed correctly — PASS

TC#	Module	Test Scenario	Expected Result	Actual / Status
		Hindi		
TC-14	NyayaSetu App	Voice query: Mera claim kahan hai (Hindi)	Nyaya Mitra responds with claim status in Hindi audio	Voice response — PASS
TC-15	Performance	100 concurrent OCR requests to FastAPI endpoint	All requests respond; P95 latency under 12 seconds	Avg: 8.4s; P95: 11.2s — PASS
TC-16	Security	SQL injection attempt in claim_id URL parameter	Request rejected with HTTP 400; incident logged in audit	Rejected and logged — PASS

Table 13: Test Cases and Results (16 Test Cases)

CHAPTER 8: CONCLUSION

8.1 Conclusion

Astitava represents a significant and measurable step forward in the digital governance of India-s Forest Rights Act claim administration. By integrating AI-powered document intelligence (Tesseract OCR + IndicBERT NER), GPS-based geospatial boundary capture and PostGIS spatial validation, Random Forest-based DSS eligibility scoring with a 17-scheme Welfare Registry, and a multilingual citizen-facing NyayaSetu mobile application with voice assistant, the platform addresses the complete FRA claim lifecycle within a unified, offline-capable digital ecosystem designed specifically for deployment in low-connectivity forest regions.

The pilot deployment across four states — Madhya Pradesh, Odisha, Chhattisgarh, and Jharkhand — processed 15,247 real FRA claims with an average processing time of 3.8 days against a 47-day manual baseline, representing a 92% reduction in claim-to-decision time. Patta issuance rates in pilot areas reached 61.4% against a 34.2% state average prior to deployment, directly increasing the number of tribal families receiving formal land rights. DSS scheme matching achieved 89.7% precision, connecting eligible beneficiaries to welfare entitlements they would not have accessed through manual processes. The NyayaSetu app achieved 8,430 installs among tribal beneficiaries in pilot areas, demonstrating grassroots adoption at scale.

The platform has received formal recognition from the Ministry of Tribal Affairs and has been recommended for scale-up to five additional states. Astitava demonstrates conclusively that targeted, domain-specific AI applications built with deep contextual understanding of India-s governance landscape, and validated against real-world field conditions, can transform the delivery of constitutional rights at scale with measurable, auditable impact.

8.2 Future Work

Six directions are identified for future development of the Astitava platform:

- Generative AI Integration: Fine-tune a domain-specific Large Language Model (LLM) on the complete FRA legal corpus — the Act, Rules, MoTA circulars, and state-level orders — to enable natural language querying of claim status, rights entitlements, and appeal procedures; replacing the current rule-based Nyaya Mitra with a conversational generative AI voice assistant capable of handling complex multi-turn rights queries in 12 Scheduled languages.
- Predictive Claim Risk Analytics: Develop a predictive model using the 15,247-claim pilot corpus to identify FRA claims at risk of SDLC rejection at the time of submission. Early warning flags would enable Van Adhikar Mitras to correct deficiencies proactively before official review, significantly improving first-submission approval rates.
- Inter-State Portability and Cross-District Claim Verification: Enable secure cross-state claim verification for tribal communities whose cultivated land spans administrative boundaries — a common occurrence for communities with traditional seasonal land-use patterns across state or district borders.

- Carbon Credit and Ecosystem Services Integration: Link FRA-certified forest land parcels to voluntary carbon credit markets (such as India-s domestic carbon market under the Carbon Credit Trading Scheme, 2023), creating a legally-grounded income stream for Patta-holding tribal communities based on forest conservation commitments.
- ABHA and DigiLocker Integration: Link Patta certificates issued through Astitava to beneficiaries-s Ayushman Bharat Health Account (ABHA) and DigiLocker digital document wallet, enabling seamless welfare delivery and document portability across all government services.
- Edge AI Deployment: Compress and quantise the Tesseract + IndicBERT pipeline using ONNX Runtime and INT8 quantisation to enable fully offline AI document processing on Android devices without cloud API calls — critical for the approximately 18% of field sessions that occur in areas with zero cellular connectivity.

REFERENCES

[1] Ministry of Tribal Affairs, Government of India, "The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006," Gazette of India Extraordinary, Part II, Section 1, 2006.

[2] Ministry of Tribal Affairs, "National Forest Rights Act Rules, 2008," Government of India, New Delhi, 2008.

[3] Sontakke, A. and Kulkarni, P., "GIS-Based Land Record Management System for Rural Maharashtra," in Proc. IEEE International Conference on ICT Applications (ICTACT), 2021, pp. 112–118.

[4] Joshi, R., Mishra, A., and Patil, K., "Optical Character Recognition for Hindi Government Documents: Benchmarks and Fine-Tuning Strategies," in Proc. EMNLP Workshop on Indic Natural Language Processing, 2022, pp. 45–52.

[5] Kakwani, D., Kunchukuttan, A., Golla, S., Gokul, N. C., Iyer, A., Khapra, M. M., and Kumar, P., "IndicNLP Suite: Monolingual Corpora, Evaluation Benchmarks and Pre-Trained Multilingual Language Models for Indian Languages," in Findings of EMNLP, arXiv:2212.05000, 2022.

[6] National Informatics Centre, "BhuNaksha GIS Platform for Land Record Digitisation," NIC Technical Report TR-2021-07, Government of India, 2021.

[7] Indian Space Research Organisation, "Bhuvan National Geoportal: Satellite Data Access for e-Governance Applications," Space Applications Centre, Ahmedabad, 2023.

[8] Ramachandran, P., Venkataraman, S., and Bhat, R., "Random Forest-Based Welfare Scheme Eligibility Scoring in Karnataka: A Decision Support Systems Case Study," Journal of e-Governance, vol. 14, no. 3, pp. 211–229, 2022.

[9] Breiman, L., "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, Kluwer Academic Publishers, 2001.

[10] PostGIS Project Steering Committee, "PostGIS 3.3 Reference Manual," OSGeo Foundation, 2023. [Online]. Available: https://postgis.net/docs/manual-3.3/

[11] Meta Open Source, "React Native 0.72 Documentation," 2023. [Online]. Available: https://reactnative.dev/

[12] Mukherjee, S. and Dasgupta, R., "Offline-First Mobile Architecture for Rural India: Challenges and Design Patterns," IEEE Access, vol. 10, pp. 23411–23424, 2022.

[13] Ministry of Electronics and Information Technology, "Digital Personal Data Protection Act, 2023," Gazette of India Extraordinary, Part II, Section 1, 2023.

[14] Ministry of Tribal Affairs, "Annual Report 2024–25: FRA Implementation Status across States," New Delhi, Government of India, 2025.

WEBSITES

[W1] Ministry of Tribal Affairs FRA Portal. Available: https://fra.org.in

[W2] Bhuvan National Geoportal, ISRO. Available: https://bhuvan.nrsc.gov.in

[W3] PostGIS Official Documentation. Available: https://postgis.net/docs/

[W4] React Native Official Documentation. Available: https://reactnative.dev/

[W5] FastAPI Framework Documentation. Available: https://fastapi.tiangolo.com/

[W6] AI4Bharat — IndicBERT and Indic NLP Resources. Available: https://ai4bharat.org/

[W7] Sentinel-2 Satellite Data Access — ESA Copernicus. Available: https://sentinel.esa.int/

[W8] AWS Documentation — EC2, RDS, S3, CloudFront. Available: https://docs.aws.amazon.com/

[W9] UMANG Unified Mobile Application, Government of India. Available: https://web.umang.gov.in/

[W10] DigiLocker Digital Document Wallet. Available: https://www.digilocker.gov.in/

[W11] Hugging Face Transformers Library. Available: https://huggingface.co/transformers/

[W12] Leaflet.js Interactive Maps Library. Available: https://leafletjs.com/

[W13] Azure Cognitive Services Speech SDK. Available: https://learn.microsoft.com/en-us/azure/cognitive-services/speech-service/

[W14] Scikit-learn Machine Learning Library. Available: https://scikit-learn.org/

[W15] SQLCipher Encrypted SQLite. Available: https://www.zetetic.net/sqlcipher/

APPENDIX

Appendix A: Project Log Note Format

The project maintained a fortnightly sprint log for each of the six development phases. Each log entry captures: Sprint Number, Sprint Start Date, Sprint End Date, Sprint Objectives, Tasks Completed (with commit references), Issues and Blockers Encountered, Resolutions Applied, Next Sprint Goals, and Guide Review Sign-off. Sprint logs were reviewed and co-signed by Project Guide Dr. Nisha Rathi and Project Coordinator Prof. Ashish Anjana at the end of each two-week cycle. A total of 16 sprint logs were maintained across the 32-week development timeline.

Sprint #	Dates	Module Delivered	Key Issues Resolved
1	Weeks 10–12	OCR/NER Engine v1	IndicBERT VRAM OOM — resolved by batch inference with size 4
2	Weeks 13–15	Astitava Mobile App v1	GPS drift compensation — implemented Kalman filter on vertex stream
3	Weeks 16–18	WebGIS + PostGIS Module v1	ST_Intersects slow on 10k polygons — resolved with GiST index
4	Weeks 19–21	DSS Engine v1	RF overfitting on small corpus — resolved with 5-fold cross-validation
5	Weeks 22–24	Web Portal + NyayaSetu v1	Leaflet tile performance on mobile — resolved with clustered markers

Table A1: Sprint Log Summary

Appendix B: Component Datasheets Summary

Component	Version	License	Role in System
Tesseract OCR	5.3.1	Apache 2.0	Text extraction from scanned FRA documents
IndicBERT	Fine-tuned checkpoint	MIT / CC BY 4.0	Named Entity Recognition for Indic government text
PostGIS	3.3 (PostgreSQL 15)	GPLv2	Spatial database for GPS polygon storage and queries
React Native	0.72 (Android)	MIT	Cross-platform field mobile application framework
FastAPI	0.103	MIT	Async REST API gateway for all backend microservices
Scikit-learn	1.3	BSD 3-Clause	Random Forest DSS eligibility engine (100 estimators)

Component	Version	License	Role in System
Leaflet.js	1.9	BSD 2-Clause	WebGIS map rendering and GeoJSON layer display
SQLCipher	4.5	BSD / Commercial	AES-256 page-level encrypted SQLite for mobile storage
Azure Speech SDK	1.31	Microsoft	Hindi/Indic TTS and STT for Nyaya Mitra voice assistant
OpenCV	4.8	Apache 2.0	Document pre-processing (deskew, threshold, crop)

Table A2: Component Datasheets Summary

Appendix C: API Endpoint Documentation

Endpoint	Method	Auth	Request Body / Params	Response
/api/v1/digitize	POST	JWT Bearer	Multipart: PDF/PNG file upload	JSON: entities, CHS score, entity_count
/api/v1/claims	POST	JWT Bearer	JSON: beneficiary_id, land_type, gps_polygon_id	JSON: claim_id, status: SUBMITTED
/api/v1/claims/{id}	GET	JWT Bearer	Path: claim_id string	JSON: full claim record + DSS output
/api/v1/claims/{id}/status	PATCH	JWT Bearer (SDLC role)	JSON: status, comments	JSON: updated claim record
/api/v1/gps	POST	JWT Bearer	JSON: vertex_array (lat/lng pairs), claim_id	JSON: gps_polygon_id, area_acres, conflict_flag
/api/v1/dss/evaluate	POST	JWT Bearer	JSON: beneficiary_id, claim_id	JSON: vulnerability_score, convergence_score, scheme_list
/api/v1/schemes	GET	JWT Bearer	Query: category (optional filter)	JSON: array of 17 scheme profiles with eligibility rules
/api/v1/nyayasetu/status	GET	Phone OTP	Query: phone_number	JSON: claim_status, scheme_recommendations, patta_url

Table A3: API Endpoint Documentation

Appendix D: Glossary of Terms

Term / Abbreviation	Full Form / Definition
FRA	Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006
OCR	Optical Character Recognition — AI-based extraction of text from scanned images
NER	Named Entity Recognition — classification of named entities

Term / Abbreviation	Full Form / Definition
	(person, location, quantity) in text
PostGIS	Open-source spatial extension to PostgreSQL providing ST_Polygon, ST_Intersects, and GiST indexing
DSS	Decision Support System — Astitava-s eligibility and welfare scheme recommendation engine
NDVI	Normalised Difference Vegetation Index — satellite measure of vegetation density from Sentinel-2 bands
SDLC	Sub-Divisional Level Committee — government body reviewing FRA claims at sub-district level
DLC	District Level Committee — government body with final authority on FRA claim approval or rejection
Patta	Land title document issued to a tribal claimant on successful FRA claim approval by DLC
PVTG	Particularly Vulnerable Tribal Group — tribal communities with additional legal protections and priority scheme access
CHS	Claim Health Score — AI-generated completeness and consistency score (0.0–1.0) for each digitised FRA claim
RBAC	Role-Based Access Control — system enforcing data access based on authenticated user role and district scope
ST	Scheduled Tribe — communities listed under Article 342 of the Constitution of India
Van Adhikar Mitra	Village-level field facilitator who assists tribal claimants in filing FRA claims using the Astitava Mobile App
GPS	Global Positioning System — satellite-based geolocation system used for land boundary capture
DPDPA	Digital Personal Data Protection Act, 2023 — India-s primary legislation governing personal data processing

Table A4: Glossary of Terms